

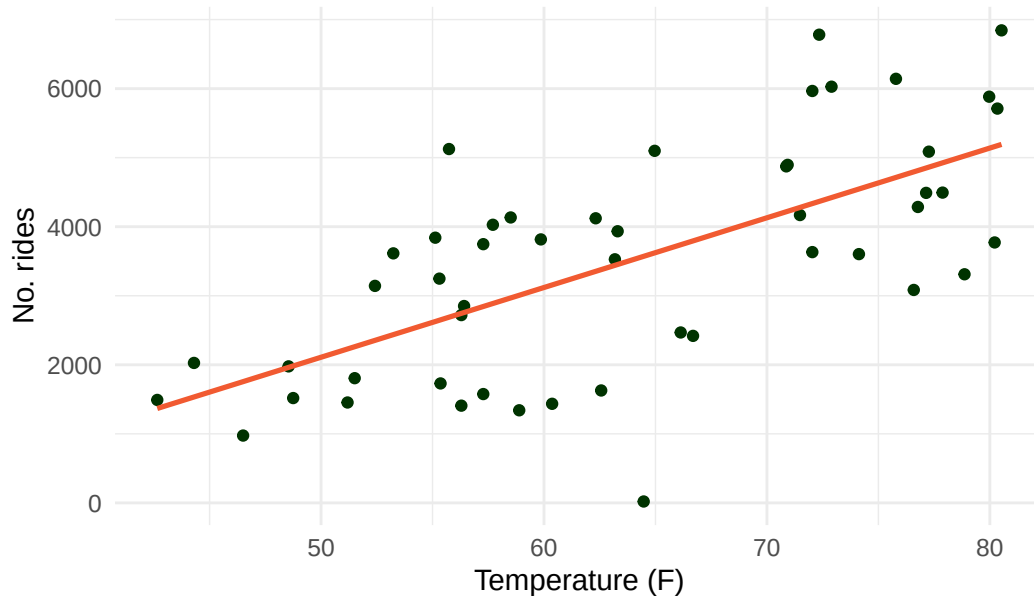
# Day 05: Evaluating SLR models

Name: \_\_\_\_\_

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## Data for today

We return to the Capital Bikeshare data, this time working with a small random sample of 50 days so that patterns are easier to see by eye.



| term        | estimate  | std. error | t     | p-value |
|-------------|-----------|------------|-------|---------|
| Intercept   | -2940.400 | 1072.078   | -2.74 | 0.009   |
| temp_actual | 100.995   | 16.567     | 6.10  | <0.001  |

## Two definitions of a "good" model

1. The model is appropriate, so our inferences (p-values, CIs) are \_\_\_\_\_
2. The x variable does a good job \_\_\_\_\_ the y variable

Theoretical model:  $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$  where  $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

Fitted model:  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ , so that  $y_i = \hat{y}_i + \hat{\varepsilon}_i$

## Conditions for inference: LINE

For our p-values and confidence intervals to be trustworthy, the theoretical model above has to be a reasonable description of the data. Check:

- **Linear:** \_\_\_\_\_
- **Independent and identically distributed errors:** \_\_\_\_\_
- **Normally distributed errors:** \_\_\_\_\_
- **Equal variance of the errors:** \_\_\_\_\_

### Note

These conditions are about the *errors*, not about the raw  $x$  or  $y$  variables themselves.

## Residuals

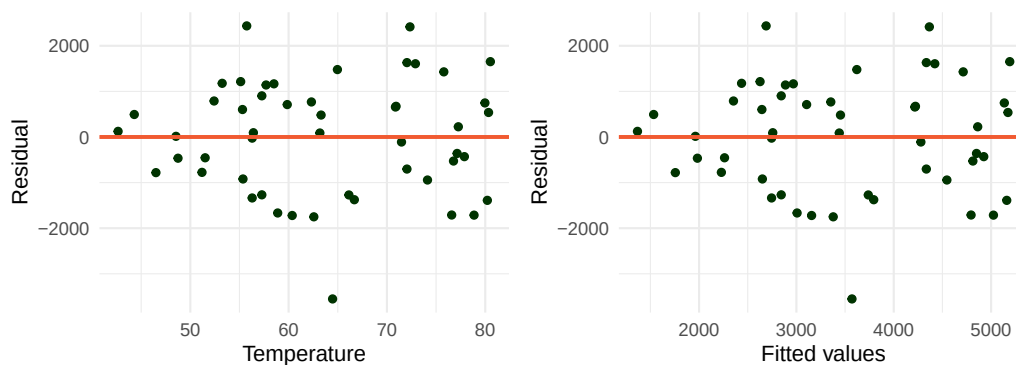
**Definition:**  $e_i = \hat{\varepsilon}_i = y_i - \hat{y}_i$

### Properties:

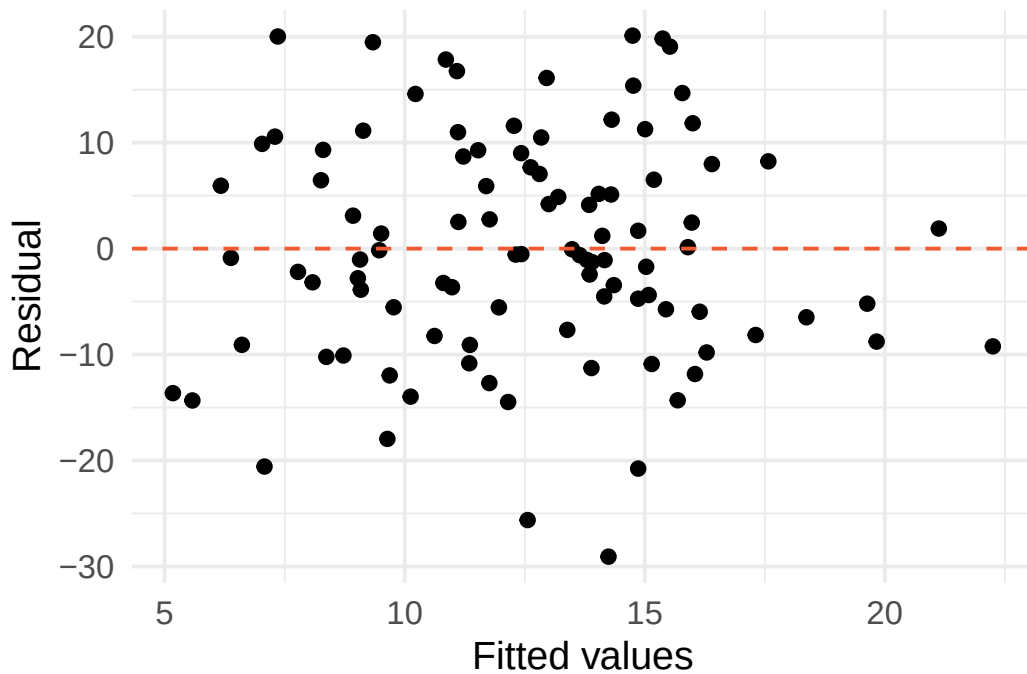
- sum to zero  $\implies$  mean is \_\_\_\_
- **uncorrelated** with  $x$  and  $\hat{y}$
- (should be) normally distributed
- $SD(e_i) = \hat{\sigma} \sqrt{1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$

## Residual plots

For SLR, plotting residuals against  $x$  or against the fitted values  $\hat{y}$  shows the same shape (since  $\hat{y}$  is a linear function of  $x$ ) – either is a valid way to check linearity and equal variance.



**A “good” residual plot** shows no pattern: residuals bounce randomly around zero with roughly constant spread across the range of fitted values.



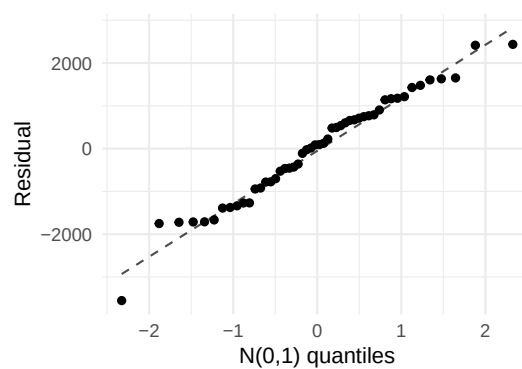
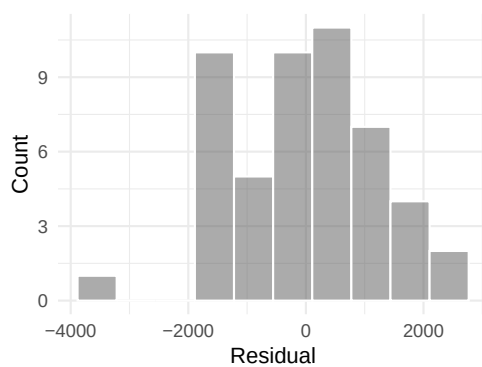
#### 💡 Group task

On the whiteboards, sketch a plot of  $y$  vs.  $x$  and the corresponding residual plot that would indicate a violation of

1. the linearity condition
2. the equal variance condition

#### Assessing normality

Use a histogram or normal Q-Q plot of the residuals. A “good” Q-Q plot hugs the diagonal reference line closely.

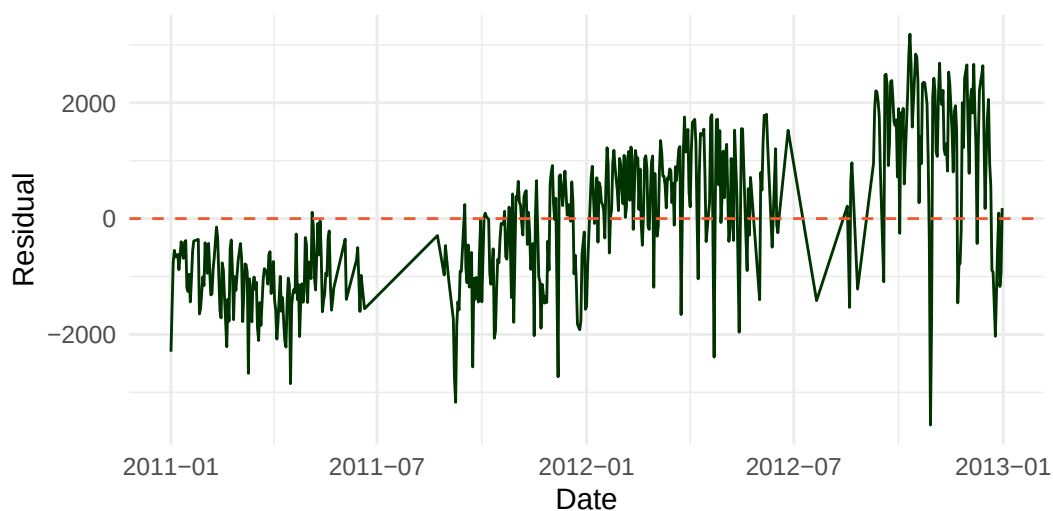


**i** Note

Normality is **harder to judge at small sample sizes** – even true random draws from a normal distribution can look lumpy or have a wandering Q-Q plot when  $n$  is small. Don't over-interpret minor wiggles with only 30-50 points.

## Assessing independence

Plot residuals against whatever variable might induce dependence (time, location, subject, batch, ...). Using the *full* bikes dataset ordered by date reveals a clear pattern – a strong hint that day-to-day ridership is **not** independent once we condition only on temperature:



What happens if the conditions aren't met?

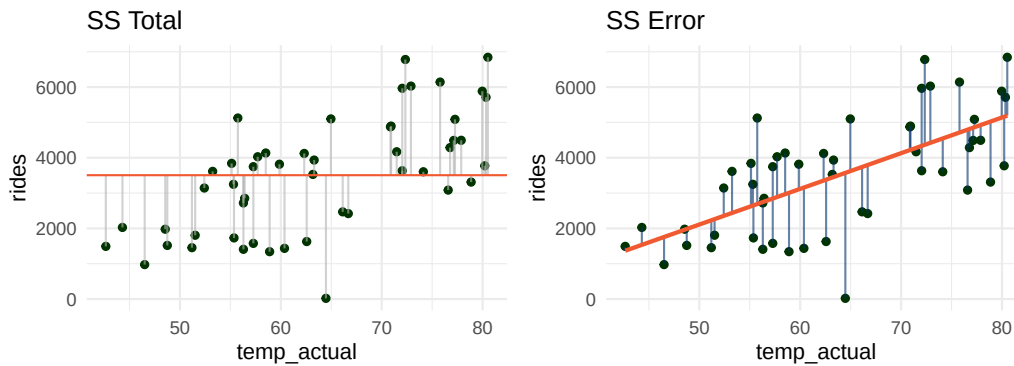
- **Linearity:** if nonlinear, everything breaks!
- **Independence:** estimates are still unbiased, but the SEs are typically too **small** (we have less information than  $n$  independent points suggests)
- **Normality:** estimates and SEs are still fine, but **prediction intervals** (which rely on individual points being normal around the mean) are wrong
- **Equal variance:** estimates are still unbiased, but SEs are wrong – and we don't know in which direction

What do we do if assumptions are violated?

1. Change our assumptions (hard – needs more advanced methods)
2. Transform  $y$ ,  $x$ , or both
3. Change the type of inference (e.g. bootstrap)

Is  $x$  a “good” predictor of  $y$ ?

Partitioning variability



$$\underbrace{\sum (y_i - \bar{y})^2}_{SSTO} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum e_i^2}_{SSE}$$

total variability = variability of  $\hat{Y}$ 's + variability of residuals (\_\_\_\_\_)

$$R^2 = \frac{SSR}{SSTO} = \frac{\text{variability of } \hat{Y}}{\text{variability of } Y}$$

**i**  $R^2$

$R^2$  gives the proportion of total variability in  $Y$  explained by the SLR model.

For our small bike sample,  $R^2 = 0.436$  – about 43.6% of the variability in rides is explained by temperature.

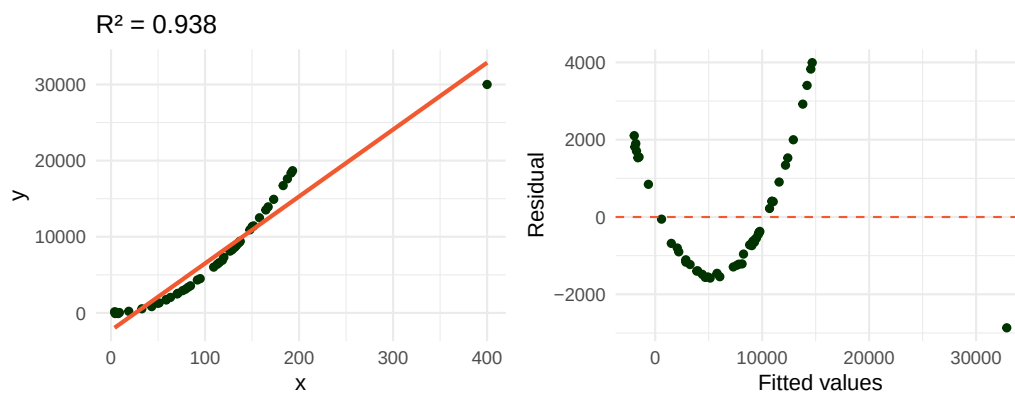
Properties of  $R^2$

- Between \_\_\_\_ and \_\_\_\_
- Big when  $\sigma$  is small relative to the overall variability of  $y$  (i.e. a big proportion of variability explained by the model)
- For SLR,  $R^2 = r^2$  (where  $r$  is the correlation coefficient between  $x$  and  $y$ )

### Quick notation reference

| Symbol    | Meaning                             |
|-----------|-------------------------------------|
| $r$       | sample correlation coefficient      |
| $\rho$    | population correlation coefficient  |
| $R^2$     | coefficient of determination        |
| $p$       | a population proportion             |
| $\hat{p}$ | a sample proportion                 |
| p-value   | $P(\text{test statistic} \mid H_0)$ |
| $P(A)$    | probability of event $A$            |

$R^2$  dangers: high  $R^2$  doesn't mean ``good fit''



A single high-leverage point drove  $R^2$  up to 0.938 even though the residual plot shows an obvious curve – the linear model is a bad fit! **Always check the residual plot, not just  $R^2$ .**

### Your turn

Would you trust this model just based on its  $R^2$ ? What does the residual plot tell you that  $R^2$  alone can't?